

PYTHON MINI PROJECT

Title: Dice Game – Player vs Computer using Python

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Subject: Fundamentals of Programming

AIM

To design and develop a Dice Game where a player competes against the computer using Python, demonstrating logic building through functions and data structures.

OBJECTIVE

- To understand logic building using functions
 - To use random number generation
 - To implement game rules using conditions
 - To use lists for storing scores
 - To develop an interactive game
 - To apply decision-making and comparison logic
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PROCEDURE (Detailed)

1. Open Python IDLE or any editor and create a file named dice_game.py.
2. Import the random module to simulate dice rolls.
3. Define a function roll_dice():
 - Generate a random number between 1 and 6
 - Return the value
4. Define a function play_round():

- Call roll_dice() for player
 - Call roll_dice() for computer
 - Compare both values
 - Return result of the round
5. Create lists:
- player_scores to store player dice values
 - computer_scores to store computer dice values
6. Ask the user for number of rounds.
7. Use a loop to:
- Play multiple rounds
 - Store results in lists
8. Display dice values for each round.
9. Compare total scores after all rounds.
10. Declare winner:
- Player wins
 - Computer wins
 - Draw
11. Display final scores and winner.
12. Test the program with multiple inputs.

PYTHON CODE

```
import random
```

```
def roll_dice():  
    return random.randint(1, 6)
```

```
def play_game():
    rounds = int(input("Enter number of rounds: "))

    player_scores = []
    computer_scores = []

    for i in range(rounds):
        print("\nRound", i + 1)

        player = roll_dice()
        computer = roll_dice()

        player_scores.append(player)
        computer_scores.append(computer)

        print("You rolled:", player)
        print("Computer rolled:", computer)

        if player > computer:
            print("You win this round!")
        elif computer > player:
            print("Computer wins this round!")
        else:
            print("It's a draw!")

    total_player = sum(player_scores)
    total_computer = sum(computer_scores)

    print("\n--- Final Result ---")
    print("Your Score:", total_player)
    print("Computer Score:", total_computer)

    if total_player > total_computer:
        print("🎉 You are the Winner!")
    elif total_computer > total_player:
        print("💻 Computer Wins!")
    else:
        print("🤝 It's a Draw!")
```

play_game()

OUTPUT

```
Enter number of rounds: 3

Round 1
You rolled: 4
Computer rolled: 2
You win this round!

Round 2
You rolled: 1
Computer rolled: 5
Computer wins this round!

Round 3
You rolled: 6
Computer rolled: 6
It's a draw!

--- Final Result ---
Your Score: 11
Computer Score: 13
Computer Wins!
```

RESULT

The Dice Game was successfully implemented using Python. The program demonstrated logic building, random number generation, and use of functions and data structures effectively.

CONCLUSION

Thus, the mini project was successfully developed using Python. It helped in understanding core programming logic, decision-making, and real-time game simulation.